

ANALYTICAL STUDY OF A COMPLEX SYSTEM WITH REGARD TO DEPENDABILITY

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ABSTRACT

In this paper an analytical study of a complex system having two types of components of class L_1 having m components and class L_2 having n components with exponential failure time distributions and general repair time distributions have been examined. The Laplace transforms of various probabilities that the system is operating under normal efficiency, reduced efficiency and is in the failed state have been obtained. Some particular cases are discussed in detail. The well-known supplementary variable techniques (Cox (1955)) are employed to obtain the solutions. In the end, the behaviour of the complex system under steady state has been examined.

Introduction

Garg (1963) has considered the problem on the behaviour of a complex system consisting of two types of components namely of class L_1 having one component only and class L_2 having n components such that a failure in L_1 only contributes toward reduction in normal efficiency of the system instead of making it completely inoperable, whereas a failure in L_2 brings about the complete breakdown of the system, with exponential failure and repair time distributions. Malhotra (1968) extended this to the case of m components in the class L_1 retaining the assumption of exponential distributions throughout. In this paper the problem is further extended by allowing more general repair time distributions for both classes. Each component in L_2 fails with constant rate λ_i ($i=1,2,\dots,n$). Failure of any component in L_2 makes the system completely inoperable and at this state no further failures are allowed either in L_1 or in L_2 until the failed component of L_2 is repaired. In class L_1 the failure of the component follows exponential distributions with constant rate λ'_j ($j=1,2,\dots,m$) and only one failure is allowed at any instant in this class. The repair time in each component follows the general distributions in both the classes, L_1 and L_2 with probability densities $G_j(x)$ and $S_i(y)$, respectively. ($j=1,2,\dots,m$) and ($i=1,2,\dots,n$).

It has been assumed that the system is allowed to change from reduced efficiency to complete breakdown, but repair of the component of L_2 implies the repair of the component of L_1 also.

Notations

We define the following probabilities:-

$P_0(t)$ $\equiv$ the probability that at time t , the system is operating in the state of normal efficiency.

$P_{O,R_j}(x,t) \Delta \equiv$ the probability that at time t , the system which is working in the state of reduced efficiency, due to the failure of the j th component in L_1 , is being repaired and the elapsed repair time lies in the interval $(x, x + \Delta)$.

$P_{F_i}(y,t) \Delta \equiv$ The probability that at time t , the system which has failed because of the failure of the i th component in L_2 is being repaired and the elapsed repair time lies in the interval $(y, y + \Delta)$.

Now $P_{O,R}(t)$ and $P_F(t)$, the probabilities that at time t , the system is working in the state of reduced efficiency due to the failure of any one of the m components in class L_1 and the system is in the failed state due to failure of any one of the n components in class L_2 are given by

$$P_{O,R}(t) = \sum_{j=1}^m \int_0^{\infty} P_{O,R_j}(x,t) dx$$

and

$$P_F(t) = \sum_{i=1}^n \int_0^{\infty} P_{F_i}(y,t) dy$$

Mathematical formulation of the model

Elementary probability considerations lead to the following forward equations for the process:-

$$(1) \quad P_O(t + \Delta) = P_O(t)(1 - \lambda \Delta)(1 - \lambda' \Delta) + \sum_{j=1}^m \int_0^{\infty} P_{O,R_j}(x,t) \phi_j(x) dx \Delta \\ + \sum_{i=1}^n \int_0^{\infty} P_{F_i}(y,t) \eta_i(y) dy \Delta + O(\Delta)$$

$$(2) \quad P_{O,R_j}(x + \Delta, t + \Delta) = P_{O,R_j}(x,t) [1 - \phi_j(x) \Delta] (1 - \lambda \Delta) + O(\Delta)$$

and

$$(3) \quad P_{F_i}(y + \Delta, t + \Delta) = P_{F_i}(y,t) [1 - \eta_i(y) \Delta] + O(\Delta)$$

$$\text{where } \lambda = \sum_{j=1}^m \lambda_j$$

$$\lambda' = \sum_{i=1}^n \lambda'_i$$

$\phi_j(x) \Delta =$ the first order probability that the j th component of class L_1 is repaired in the interval between x and $x + \Delta$, on condition that it has been working in the state of reduced efficiency up to time x .

and

$\eta_i(y) \Delta =$ the first order probability that the i th component of class L_2 is repaired in the interval between y and $y + \Delta$, on condition that the system has not been repaired up to time y .

The relations between $\phi_j(x)$ and the probability density $G_j(x)$ and $\eta_i(y)$ and the probability density $S_i(y)$ are given by

$$(4) \quad G_j(x) = \phi_j(x) \exp \left[- \int_0^x \phi_j(x) dx \right]$$

and

$$(5) \quad S_i(y) = \eta_i(y) \exp \left[- \int_0^y \eta_i(y) dy \right]$$

a detailed description of which appears in Davis (1952) in connection with failure time distributions.

Equations (1) to (3) as $\Delta \rightarrow 0$ become,

$$(6) \quad \left[\frac{\delta}{\delta t} + \lambda + \lambda' \right] P_o(t) = \sum_{j=1}^m \int_0^{\infty} P_{o,R_j}(x,t) \phi_j(x) dx \\ + \sum_{i=1}^n \int_0^{\infty} P_{F_i}(y,t) \eta_i(y) dy$$

$$(7) \quad \left[\frac{\delta}{\delta t} + \frac{\delta}{\delta x} + \lambda + \phi_j(x) \right] P_{o,R_j}(x,t) = 0$$

and

$$(8) \quad \left[\frac{\delta}{\delta t} + \frac{\delta}{\delta y} + \eta_i(y) \right] P_{F_i}(y,t) = 0$$

Equations (6) to (8) are to be solved under the boundary conditions:

$$(9) \quad P_{o,R_j}(0,t) = \lambda'_j P_o(t)$$

and

$$(10) \quad P_{F_i}(0,t) = \lambda_i [P_o(t) + P_{o,R}(t)]$$

which arise from consideration of the rates at which failures occur.

Solution of the Model

Assume that the system is operating in the state of normal efficiency initially, so that $P_o(0) = 1$, and $P_{o,R_j}(x,0) = P_{F_i}(y,0) = 0$ for $x, y > 0$.

Let the Laplace transform of the function $f(t)$ be denoted by $\bar{f}(s)$, i.e.

$$\bar{f}(s) = \int_0^{\infty} e^{-st} f(t) dt \quad \text{Re}(s) > 0$$

Applying the Laplace transforms and making use of the initial conditions mentioned above, equations (6) to (10) become

$$(11) \quad (s + \lambda + \lambda') \bar{P}_o(s) = 1 + \sum_{j=1}^m \int_0^{\infty} \bar{P}_{o,R_j}(x,s) \phi_j(x) dx \\ + \sum_{i=1}^n \int_0^{\infty} \bar{P}_{F_i}(y,s) \eta_i(y) dy$$

$$(12) \quad \left[\frac{\delta}{\delta x} + s + \lambda + \phi_j(x) \right] \bar{P}_{o,R_j}(x,s) = 0$$

$$(13) \quad \left[\frac{\delta}{\delta y} + s + \eta_i(y) \right] \bar{P}_{F_i}(y, s) = 0$$

$$(14) \quad \bar{P}_{O, R_j}(0, s) = \lambda'_j \bar{P}_O(s)$$

and

$$(15) \quad \bar{P}_{F_i}(0, s) = \lambda_i [\bar{P}_O(s) + \bar{P}_{O, R}(s)]$$

Equations (12) and (13) on integration and simplification give

$$(16) \quad \bar{P}_{O, R_j}(x, s) = \bar{P}_{O, R_j}(0, s) \exp[-(s + \lambda)x] \exp\left[-\int_0^x \phi_j(x) dx\right]$$

and

$$(17) \quad \bar{P}_{F_i}(y, s) = \bar{P}_{F_i}(0, s) \exp(-sy) \exp\left[-\int_0^y \eta_i(y) dy\right]$$

Now $\bar{P}_{O, R}(s)$, the Laplace transform of the probability $P_{O, R}(t)$ that the system is working in reduced efficiency due to the failure of any one of the m components in L_1 is given by

$$\bar{P}_{O, R}(s) = \sum_{j=1}^m \int_0^{\infty} \bar{P}_{O, R_j}(x, s) dx$$

Making use of the relations (14) and (16) in the above expression and integrating by parts we get

$$(18) \quad \bar{P}_{O, R}(s) = A(s + \lambda) \bar{P}_O(s)$$

$$\text{where } A(s) = s^{-1} \sum_{j=1}^m \lambda'_j [1 - \bar{G}_j(s)]$$

Now substituting the values of $\bar{P}_{O, R_j}(x, s)$ and $\bar{P}_{F_i}(y, s)$ from (16) and (17) and using (14), (15) and (18) in (11) and solving for $\bar{P}_O(s)$ we get

$$(19) \quad \bar{P}_O(s) = 1/s [1 + A(s + \lambda)] [1 + B(s)]$$

$$\text{where } B(s) = s^{-1} \sum_{i=1}^n \lambda_i [1 - \bar{S}_i(s)]$$

Substituting this value of $\bar{P}_O(s)$ in (18) we have

$$(20) \quad \bar{P}_{O, R}(s) = A(s + \lambda)/s [1 + A(s + \lambda)] [1 + B(s)]$$

Also $\bar{P}_F(s)$, the Laplace transform of the probability $P_F(t)$ that the system is under repair due to failure of the i th component in L_2 is given by

$$\bar{P}_F(s) = \sum_{i=1}^n \int_0^{\infty} \bar{P}_{F_i}(y, s) dy$$

which with the help of relations (15), (17), (19) and (20) and integrating by parts becomes

$$(21) \quad P_F(s) = B(s)/s[1 + B(s)]$$

Relations (19), (20) and (21) can be inverted for given values of $G_j(s+\lambda)$ and $S_i(s)$.

Behaviour of the Complex System under Steady-state

Making use of the corollary of Abel's theorem in Laplace transform (Widder (1941)) viz:

$$\lim_{s \rightarrow 0} sf(s) = \lim_{t \rightarrow \infty} f(t)$$

(if the limit on the right exists), P_O , $P_{O,R}$ and P_F (that the system is operating under normal efficiency, reduced efficiency or in the failed state respectively, when the state of statistical equilibrium has been reached) may be obtained for the general case from relations (19), (20) and (21) as

$$(22) \quad P_O = (1 + A(\lambda))^{-1} (1 + c)^{-1}$$

$$(23) \quad P_{O,R} = A(\lambda) (1 + A(\lambda))^{-1} (1 + c)^{-1}$$

$$(24) \quad P_F = c(1 + c)^{-1}$$

where $c = \sum_{i=1}^n \lambda_i / \ell_i$ with $1/\ell_i$ the mean repair time for the i th component in L_2 under very weak condition on $S_i(y)$, i.e., the existence of the mean of the repair time distribution.

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REFERENCES

- COX, D.R. (1955). The analysis of non-Markovian stochastic Processes by supplementary variables. *Proc. Camb. Phil. Soc.* **51**, 433-441.
- DAVIS, D.J. (1952). An analysis of some failure data. *J. Am. Stat. Assoc.* **47**, 113-150.
- GARG, R.C. (1963). Analytical study of a complex system having two types of components. *Naval Research Logistics Quart.* **10**, 263-269.
- HOSFORD, J.E. (1960). Measures of dependability. *Oper. Res.* **8**, 53-64.
- KEILSON and KOOHARIAN. (1960). On time dependent queuing processes *Ann. Math. Statist.* **31**, 104-112.
- MALHOTRA, N.K. (1968). Operational readiness of a complex system having two types of components. *Trabajos de Estadística.* **XIX**, 117.
- MOHAN, C., GARG, R.C. and SANGAL, P.P. (1962). Dependability of a Complex System. *Oper. Res.* **10**, 310-313.
- WIDDER, D.V. (1941). The Laplace Transform. *Princeton University Press, Princeton.*